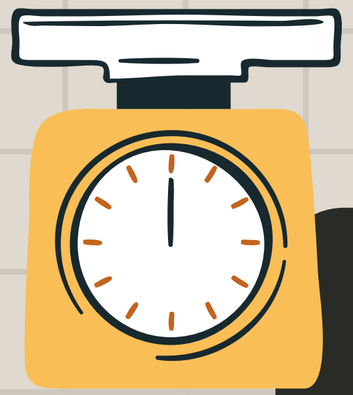
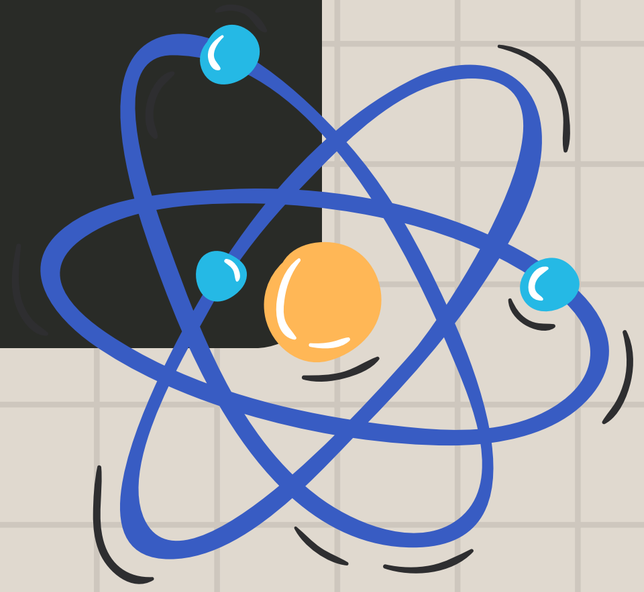


DENSITY DECODED

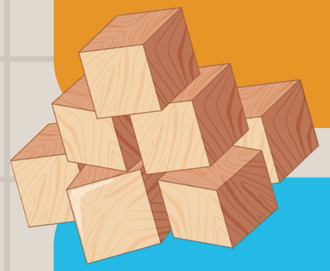
密不可分



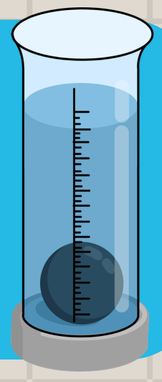
Rundown 今日任務

PART 1: WEIGHING AND CALCULATING
任務一：磅重與計算

PART 2: FINDING THE MYSTERIOUS COMBINATION
任務二：尋找組合之秘



PART 3: SHOWCASING THE ANSWER
任務三：展示答案



PART 1: WEIGHING AND CALCULATING
任務一：磅重與計算

- There are 6 different blocks
- 6個不同的物件

- Density is provided
- 會提供「密度」

- Your goal:
- 任務目標：
  - Weigh the blocks
  - 磅好物件的重量
  - Calculate the volume
  - 計算他們的體積
  - Calculator OK!
  - 可使用計數機

PART 1: WEIGHING AND CALCULATING
任務一：磅重與計算

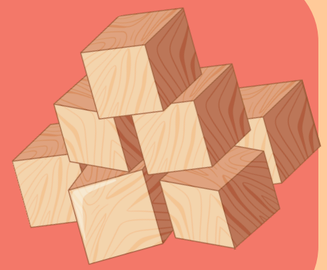
- ONLY one person may leave their seat from each group to go to the bench to weigh the blocks.
- 同一時間祇有一個組員可以誰開座位前往磅重

- The other students should use the data to calculate the volume of each block
- 其他組員應在座位計算物件的體積

- Calculator can be used.可使用計數機

- 45 Minutes later, the worksheet will be collected.
- 45分鐘後將會收集工作紙

PART 2: FINDING THE MYSTERIOUS COMBINATION
任務二：尋找組合之秘

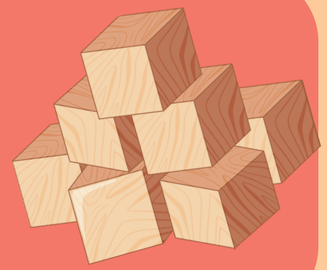


- A certain volume will be given on the screen
- 熒幕上會顯示一個隨機的體積

- Your goal: Figure out which combination can lead to that volume.
- 任務目標：組合物件，以達到目標體積。

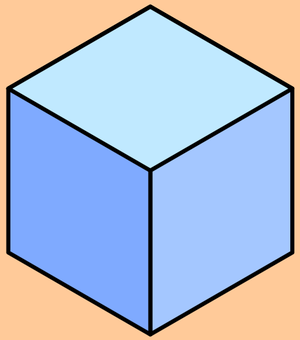
- **Special rules 特別規則：
  - One block MUST BE used TWICE 其中一個物件須使用兩次
  - Other blocks are used ONCE or ZERO times
  - 其餘物件可使用零至一次

PART 2: FINDING THE MYSTERIOUS COMBINATION
任務二：尋找組合之秘



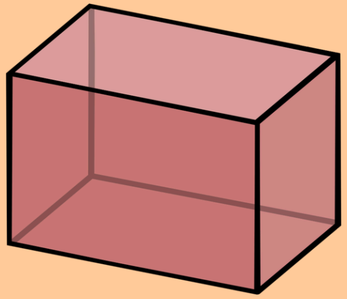
Example例子

Target目標: 30 cm^3 立方厘米



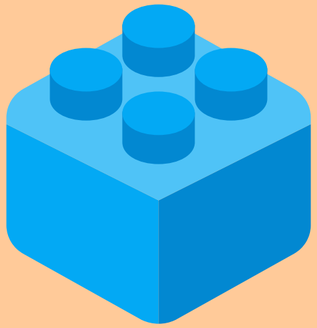
12cm^3

12 立方厘米



10cm^3

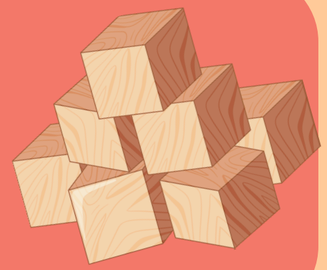
10立方厘米



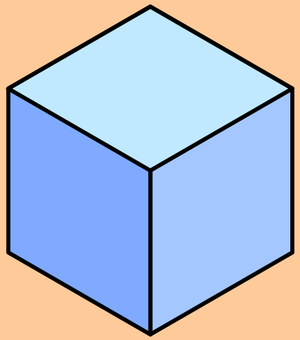
6cm^3

6立方厘米

PART 2: FINDING THE MYSTERIOUS COMBINATION
任務二：尋找組合之秘



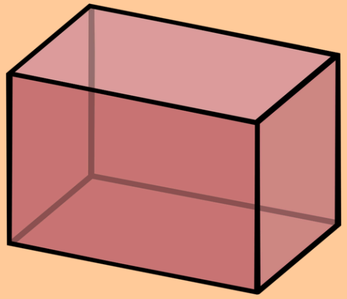
Example例子
Target目標: 30 cm^3 立方厘米



12cm^3

12 立方厘米

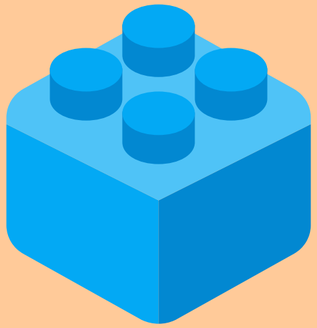
x2



10cm^3

10立方厘米

x0

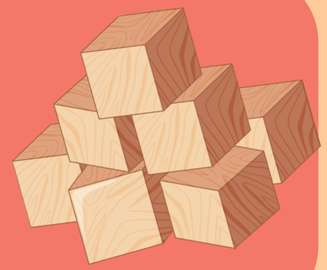


6cm^3

6立方厘米

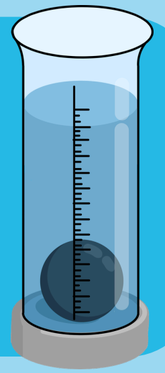
x1

PART 2: FINDING THE MYSTERIOUS COMBINATION
任務二：尋找組合之秘



- **Special rule 特別規則：
  - One block MUST BE used TWICE 其中一個物件須使用兩次
  - Other blocks are used ONCE or ZERO times
  - 其餘物件可使用零至一次
- There are MULTIPLE ANSWERS, no need to copy others!
- 答案或會多於一項，無需抄襲他人
- The worksheet will be collected after 45 minutes.
- 45分鐘後將會收集工作紙

PART 3: SHOWCASING THE ANSWER
任務三：展示答案



- Water displacement will be used to demonstrate ALL the correct answers.
- 會透過排水法，直觀地展示及驗證全部答案

MARK CALCULATIONS
計分方法

PART 2 RESULT
任務二的成績

- Groups who could get the combination correct is prioritized
- 優先計算任務二的成績

PART 1 RESULT
任務一的成績

- If there are multiple groups with correct combination, Part 1 results will be considered.
- 若任務二成績相同，則會計算任務一的成績。